

CanvasChain: Blockchain Based Authentication/Tracking for Fine Art Paintings

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Background

Art Related Crime

- FBI Art Crime Team has recovered more than 20,000 art crime items valued at over 1 billion USD
- In 2018, The Terrus museum discovered more than half its collection was forged which directed to \$200,000 financial losses

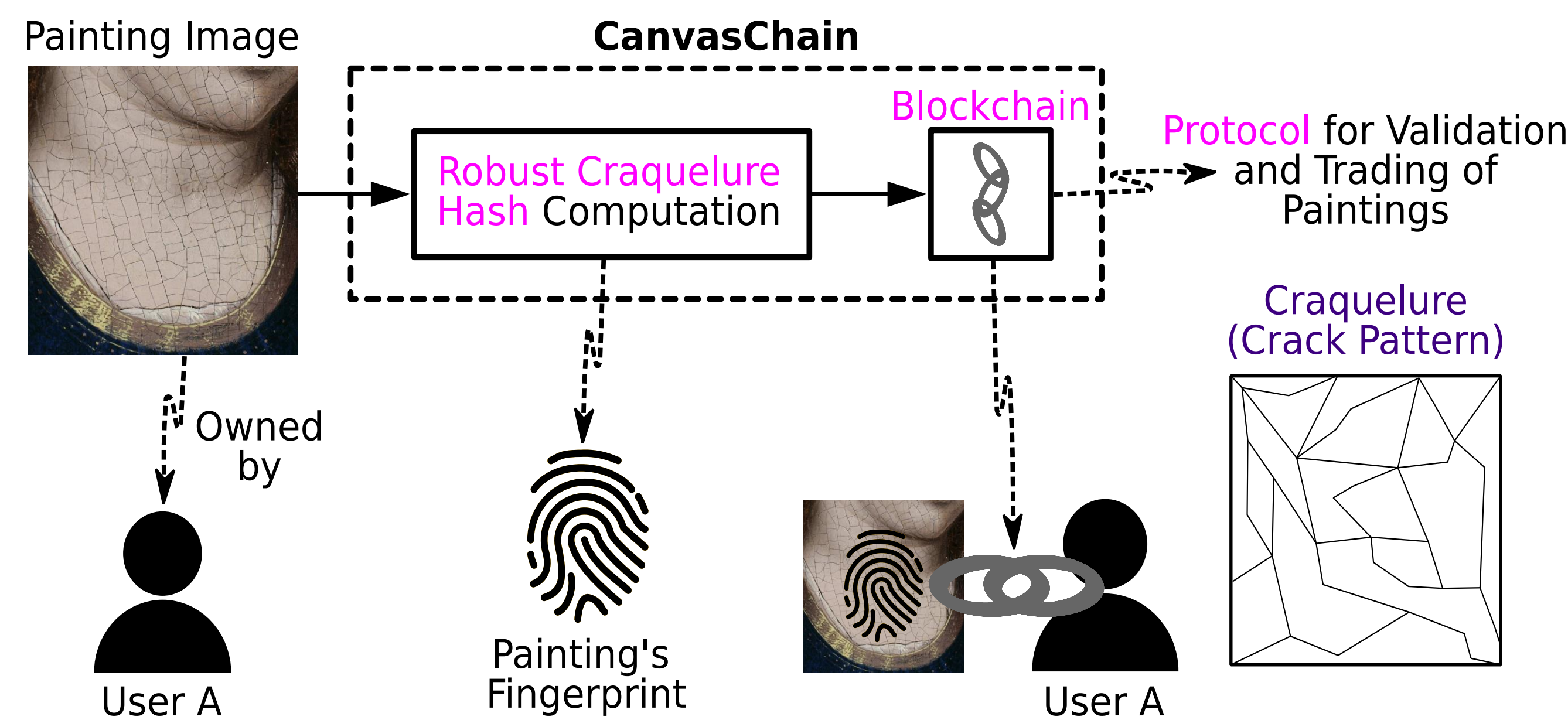
Existing Art Authentication

- Traditional authentication services usually take a long time and are usually costly
 - Often used **physical provenance** for this process (documents of the artwork's history)
- Blockchain based solutions that provide provenance do not offer a direct connection with the physical artwork

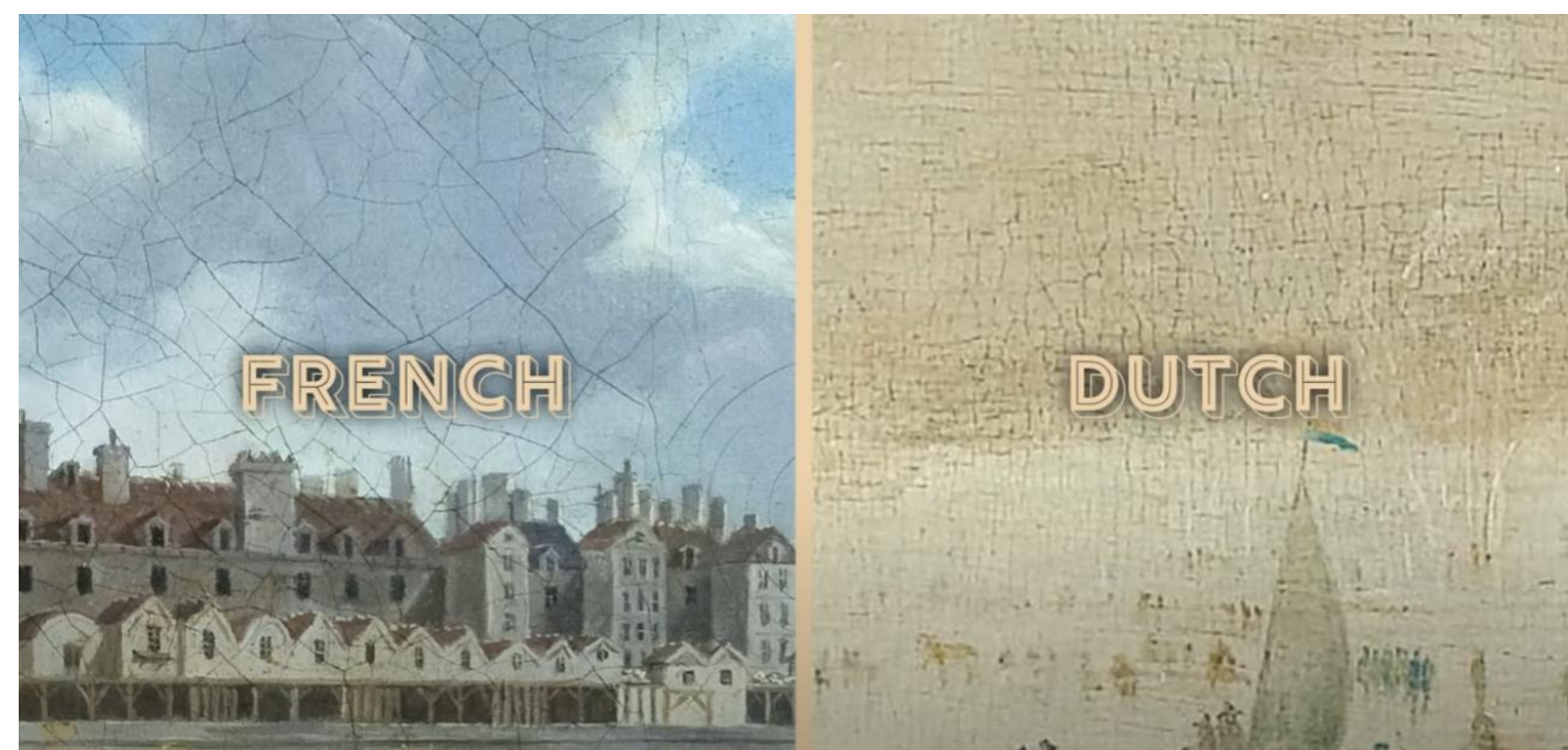


A holographic certificate from VERISART

CanvasChain Concept



- Craquelure is a unique network of cracks on the surface of paintings
 - Act as unique **"fingerprint"** of each artwork



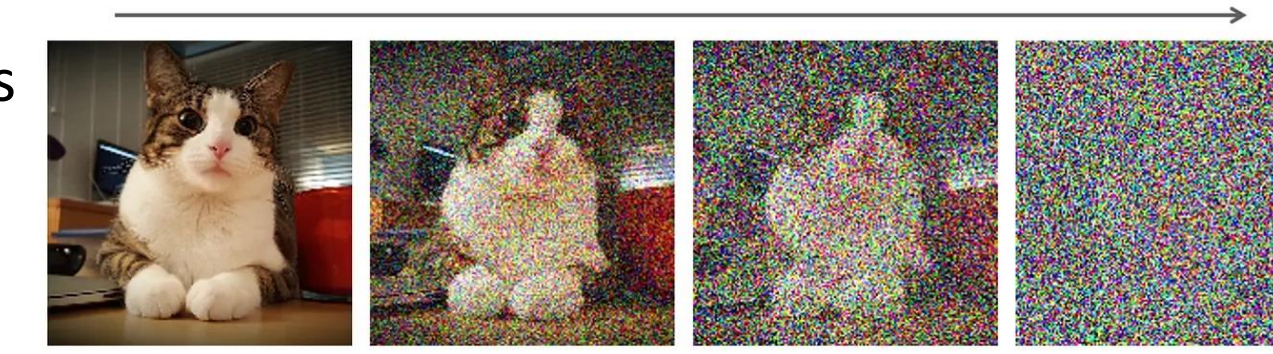
From Proof of Concept to Robust Implementation

AI Generated Craquelure Dataset

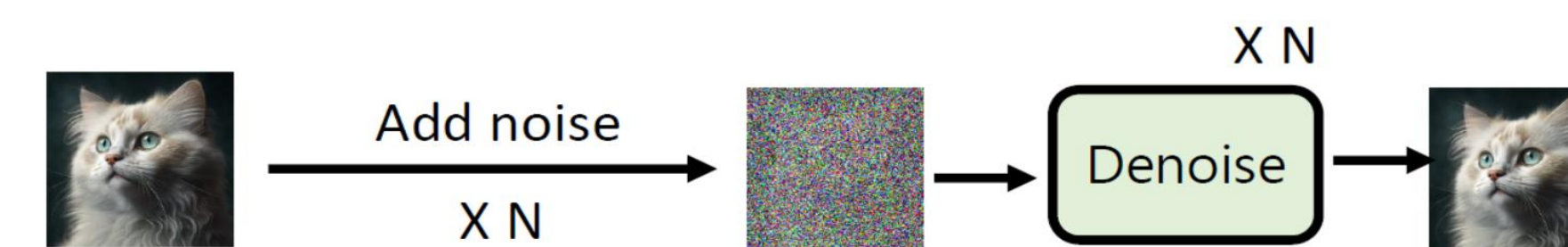
- Public craquelure datasets are scarce
- We are using AI to generate synthetic craquelure images
- Enables training of AI craquelure segmentation

Approach

- Develop a pipeline that generates synthetic craquelure images
- Based on **diffusion models** and **LoRA** (low-rank adaptation) fine tuning
- Low VRAM requirement and low cost



Noising and denoising process of diffusion model

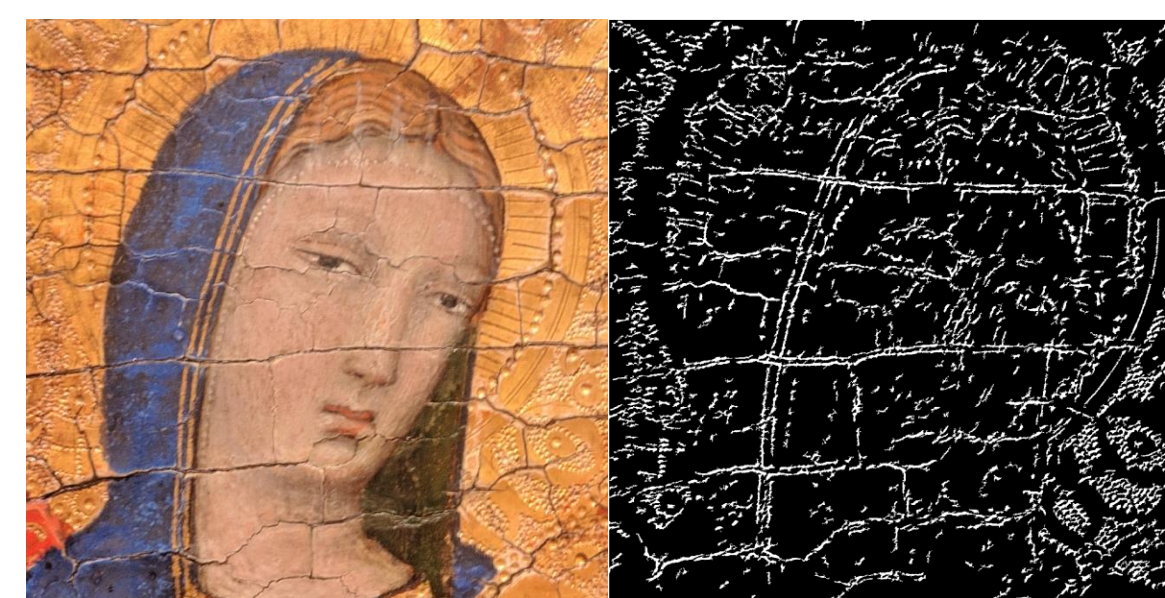


Diffusion Model

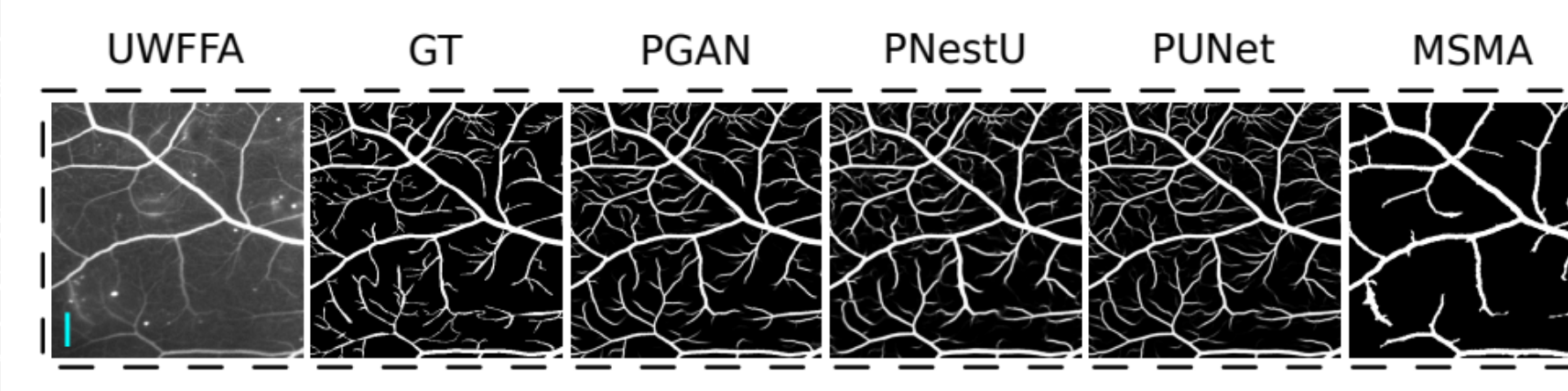
- During training, it learns how data is progressively corrupted into noise
- During generation, it denoises step by step to recover realistic samples

Craquelure Detection and Extraction

- Traditional detection methods are sensitive to **noise**
- Neural networks in machine learning have demonstrated remarkable performance in **fine object segmentation tasks**, such as cells, vessels, and craquelure



Traditional image processing method for craquelure extraction

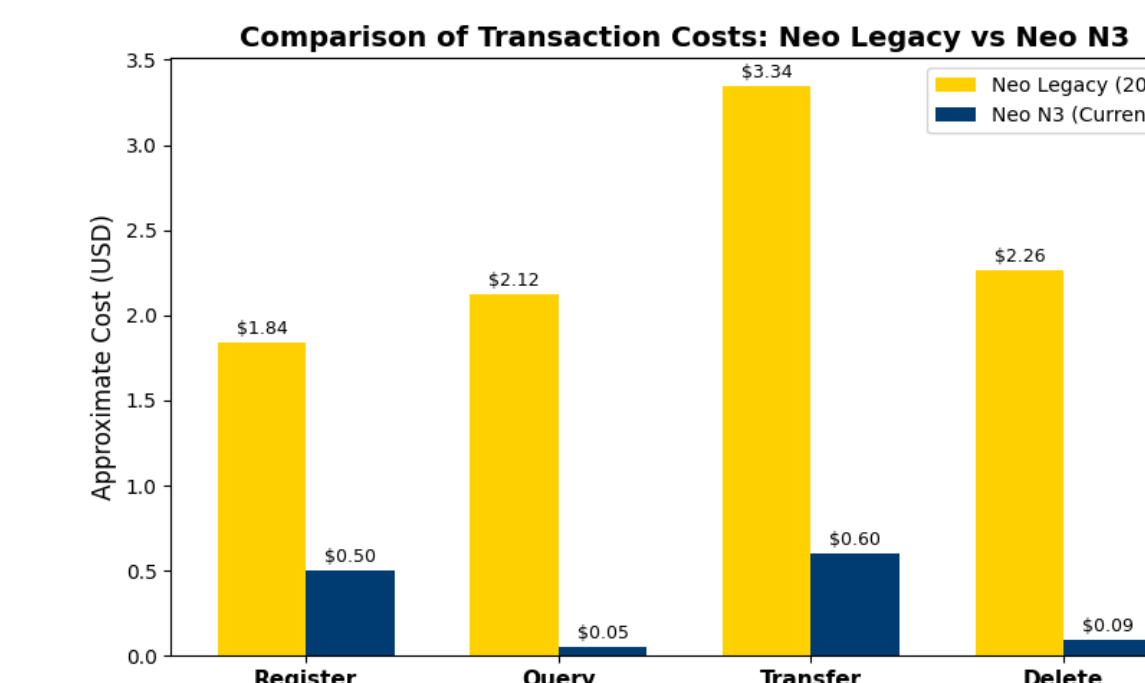


Qualitative comparison of results obtained with different neuron networks for vessel images

Blockchain Update



Neo Blockchain Updated Chain



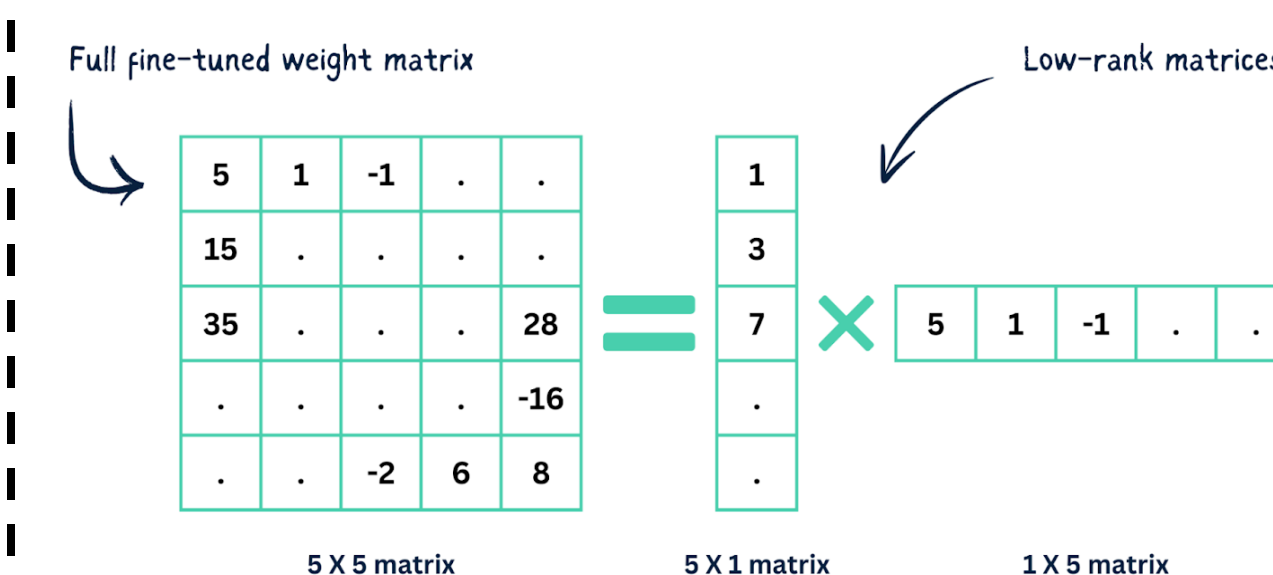
Approximate Transaction Cost of System based on Neo Legacy vs Neo N3

- CanvasChain was built on the now-deprecated NEO Legacy blockchain
- NEO N3 provides better computation capability and lower transaction cost compared to NEO Legacy
- neoFS offers decentralized storage and works with NEO N3 smart contracts

Ongoing Work

Training Workflow

- Preparing a set of craquelure images
- Use **Fluxgym** to train each LoRA checkpoint with different image caption and settings
- Build a Flux LoRA workflow in **ComfyUI**, use LoRAs to generate craquelure images
- Try different parameters and weights and pick the best-performing LoRA

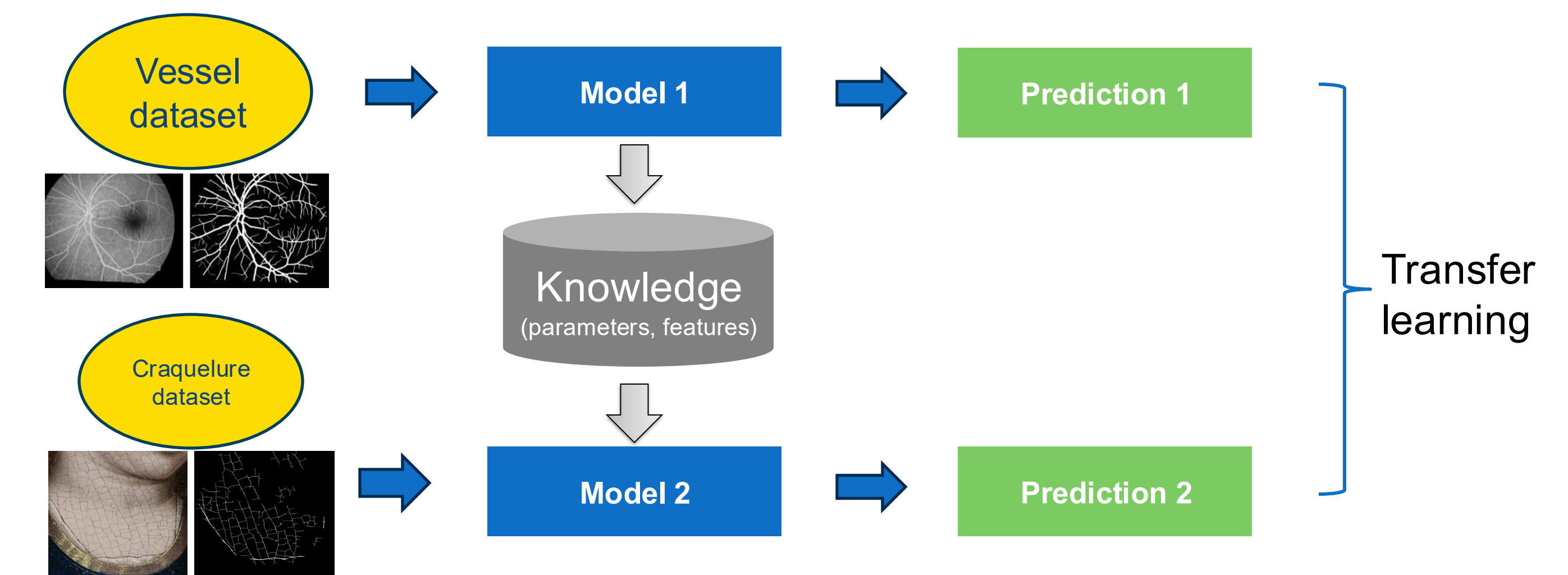


An example of low-rank matrix decomposition

LoRA

- Adds small trainable adapters
- Efficient adaptation with minimal parameters and memory use

Learning from Experience



- Reduce** the computational resources for building models

Blockchain Deployment



Simplified Smart Contract Process



Distributed Decentralized Object Storage Network

- Re-architecting the core CanvasChain smart contract for the NEO N3 blockchain
- Developing a more robust authentication method and exploring the off-chain computation options
- Building and integrating an artwork storage solution leveraging neoFS

References

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